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SIMATS ENGINEERING ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO4: Implement and evaluate network protocols and services using simulation tools to analyze performance in real-world scenarios. (BL3)

Assessment Tool Weightage: 30%

QUESTIONS: 5

Total Marks:50

Annexure A

Industry Problem Solving Task

Code of Conduct:

I Rishiykesh K.S (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

K.S.Rishiykesh

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CO4 ASSESSMENT TOOL 2

INDUSTRY PROBLEM-SOLVING TASK

Course: COMPUTER NETWORKS | Code: CSA07

Details	
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Question 1 – Star Topology for a 20-Person Startup

A star topology is ideal for a 20-person startup because all devices connect to a central switch, making the network easy to manage and troubleshoot. At the core sits a Layer 2/Layer 3 managed switch (such as a Cisco Catalyst or Netgear ProSafe) with at least 24 ports to accommodate all workstations, printers, and VoIP phones with room to grow. Each device connects to the switch using Cat6 UTP cables, which support Gigabit speeds up to 100 meters – more than sufficient for any typical office floor plan. The switch uplinks to a router/firewall (such as a Cisco RV series or pfSense appliance) that handles internet access, DHCP, and network address translation. A Wi-Fi access point connected to the switch extends wireless coverage for laptops and mobile devices. The key advantage of this design is fault isolation: if one workstation's cable fails, only that single node goes offline, leaving all other 19 employees unaffected. Centralised cable management also simplifies future moves, adds, and changes.

Problem context: A 20-person startup needs a manageable office network supporting workstations, printers, VoIP phones and wireless devices. The design must be easy to troubleshoot and allow future expansion.

Problem Analysis

- A centralized topology is appropriate because the office is relatively small and requires simple administration.
- A minimum 24-port managed switch provides ports for current devices with spare capacity for growth.
- Fault isolation is important: failure of one endpoint cable should affect only that endpoint.
- Gigabit Ethernet over Cat6 UTP is suitable for normal office distances.

Proposed Solution Design

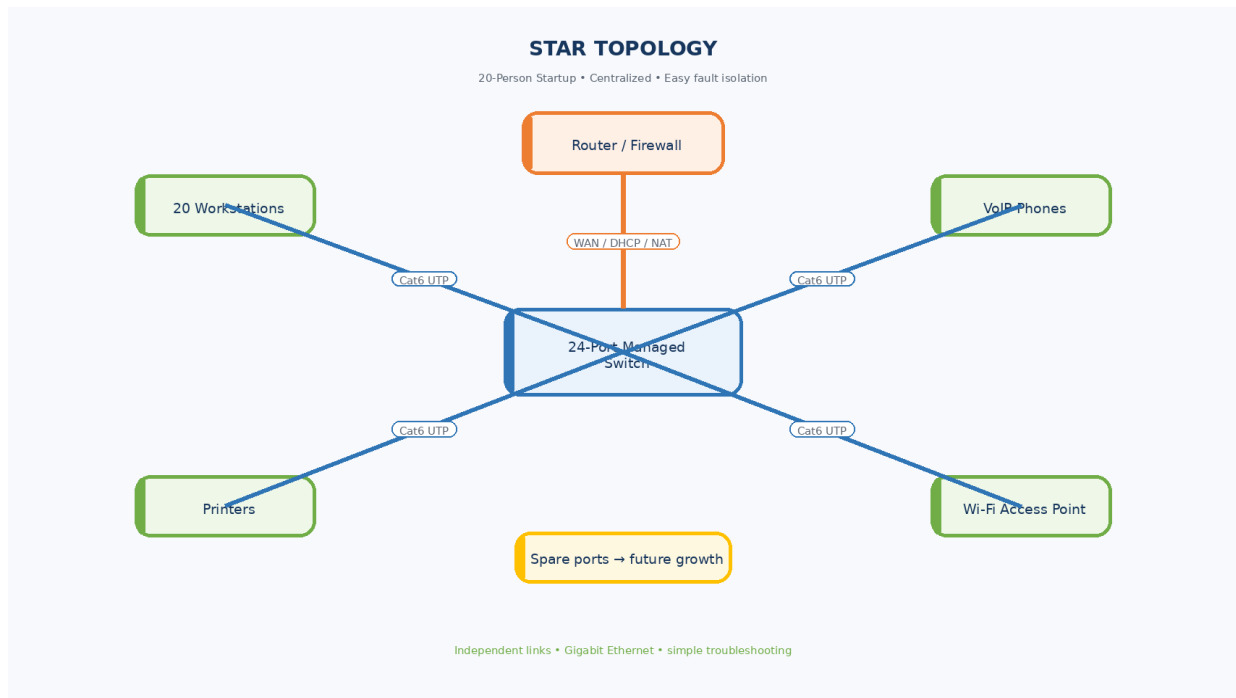


Figure 1: Star topology for the 20-person startup

- Central device: 24-port Layer 2/Layer 3 managed switch.
- Transmission medium: Cat6 UTP for endpoint connections; the source specifies support for Gigabit speeds up to 100 m.
- Edge device: router/firewall for Internet access, DHCP and NAT.
- Wireless: Wi-Fi access point connected to the switch.

- Endpoints: PCs, printers and VoIP phones connected in a star arrangement.

Technical Implementation

1. Install the managed switch at the network core of the office.
2. Connect each workstation, printer and VoIP phone directly to a switch port using Cat6 UTP.
3. Connect the switch uplink to the router/firewall.
4. Connect the Wi-Fi access point to the switch for wireless coverage.
5. Label and organize cables so moves, adds and changes can be performed efficiently.

Testing & Validation

- Disconnect one endpoint cable and verify that only that endpoint loses connectivity.
- Reconnect the endpoint and verify normal communication.
- Test Internet access through the router/firewall.
- Add a new endpoint using a spare switch port to confirm scalability.

Expected result: A centralized, scalable and easy-to-troubleshoot office network with strong fault isolation.

Question 2 — Mesh Topology for a Bank

A bank demands near-zero downtime, making a full or partial mesh topology the right choice. In a full mesh, every core network device — routers, core switches, and servers — connects directly to every other device, so no single link or node failure can bring down the network. For a bank, a partial mesh is typically deployed at the core layer (connecting all core switches and firewalls to each other) while the distribution and access layers use redundant uplinks to at least two core switches. Dual enterprise-grade routers (such as Cisco ASR or Juniper MX series) provide BGP-based redundant internet connectivity. All links between core devices use fiber optic cables — either single-mode for long distances between buildings or multi-mode OM4 within the data center — because fiber is immune to electromagnetic interference, supports very high bandwidth, and is essential for maintaining low-latency financial transactions. Redundant power supplies, UPS systems, and RAID storage arrays further complement the mesh design. The justification is straightforward: when a link fails in a mesh, traffic automatically reroutes through alternative paths in milliseconds using protocols like OSPF or BGP, satisfying regulatory requirements for uptime and data integrity.

Problem context: The bank requires near-zero downtime, high availability and reliable financial communication. The source identifies a full or partial mesh as the appropriate design.

Problem Analysis

- Availability is the primary requirement; a single failed link or node should not bring down the network.
- Core switches, routers, firewalls and servers need redundant paths.
- High bandwidth and low latency are required between critical core devices.
- The design should support automatic rerouting when a path fails.

Proposed Solution Design

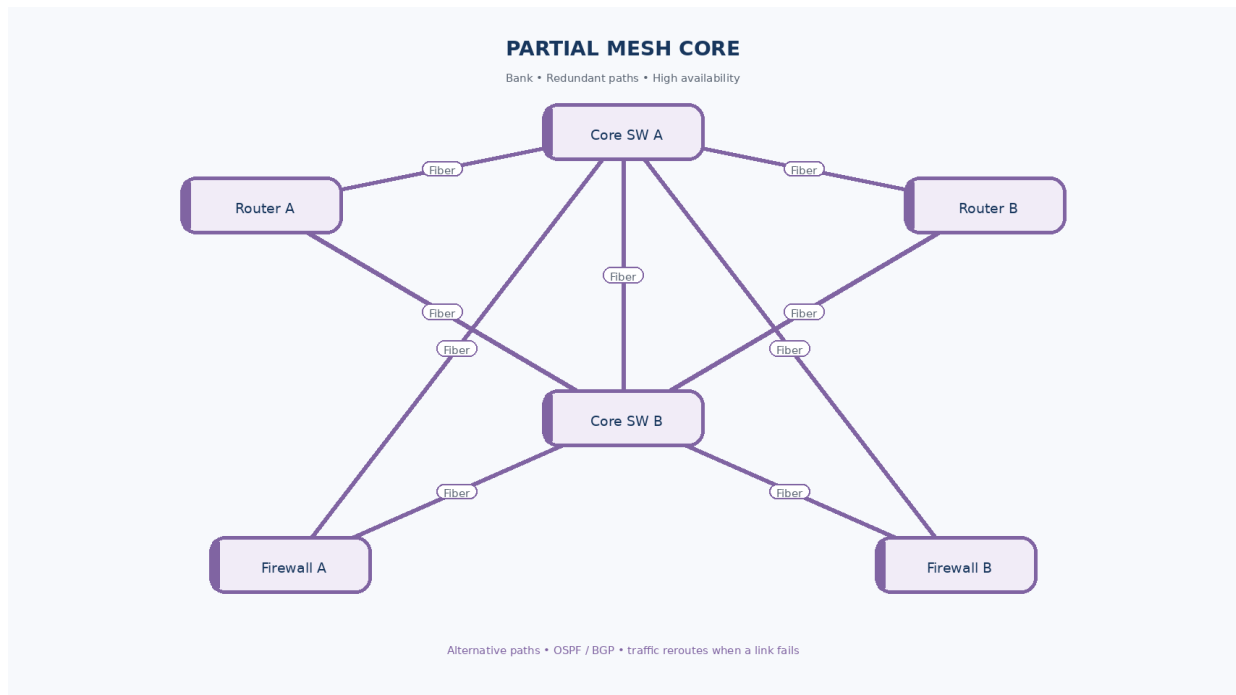


Figure 2: Partial mesh topology for the bank

- Use a partial mesh at the core so critical devices have multiple paths.
- Use redundant uplinks from distribution/access layers to at least two core switches.
- Deploy dual enterprise-grade routers for redundant Internet connectivity.

- Use fiber optic links between core devices; the source specifies single-mode for long distances and OM4 multi-mode within the data center.
- Use OSPF or BGP to support dynamic routing and rapid rerouting.
- Complement the topology with redundant power supplies, UPS systems and RAID storage.

Technical Implementation

1. Interconnect core routers, switches, firewalls and servers with redundant fiber paths.
2. Configure dynamic routing such as OSPF for internal routing and BGP for redundant Internet connectivity.
3. Provide dual uplinks from distribution/access infrastructure.
4. Validate alternate paths before production deployment.
5. Document failure domains and recovery paths.

Testing & Validation

- Disconnect one core link and verify traffic continues through an alternate path.
- Fail one router path and verify the second Internet path remains available.
- Check routing convergence and reachability after link failure.
- Verify critical servers remain accessible during a single-link failure.

Expected result: A resilient bank core in which traffic can reroute through alternative paths, reducing the effect of individual failures.

Question 3 — Bus Topology for a Small Classroom Lab

For a small classroom lab with a tight budget, bus topology offers the lowest hardware cost because all computers share a single common communication line — the bus — eliminating the need for a central switch or hub. A single coaxial cable (RG-58, 50-ohm) or, in a modern variation, a single Cat5e/Cat6 cable daisy-chained through an inexpensive unmanaged hub runs along the length of the room. Each computer connects to the backbone using a T-connector and a BNC connector in a classic coaxial setup, with terminators (50-ohm) placed at both ends of the cable to absorb signals and prevent reflection. For a more contemporary low-cost implementation, a single inexpensive 8- or 16-port unmanaged switch replaces the terminators and uses Cat5e UTP patch cables to each workstation — achieving bus-like simplicity at minimal cost. The primary limitation is that the network is susceptible to collisions and a single cable break disrupts all connected machines, but for a supervised classroom environment with light usage (internet browsing, file sharing), this trade-off is entirely acceptable. No dedicated server is required; a teacher's PC can share resources using built-in Windows/Linux file sharing.

Problem context: A small classroom lab has a tight budget and light network usage. The source proposes bus topology as the low-cost option because devices share a common communication line.

Problem Analysis

- Budget is the main constraint.
- The lab has relatively light usage such as browsing and file sharing.
- A classic bus minimizes central networking hardware.
- The major risks are collisions and the possibility that a backbone break affects all connected machines.

Proposed Solution Design

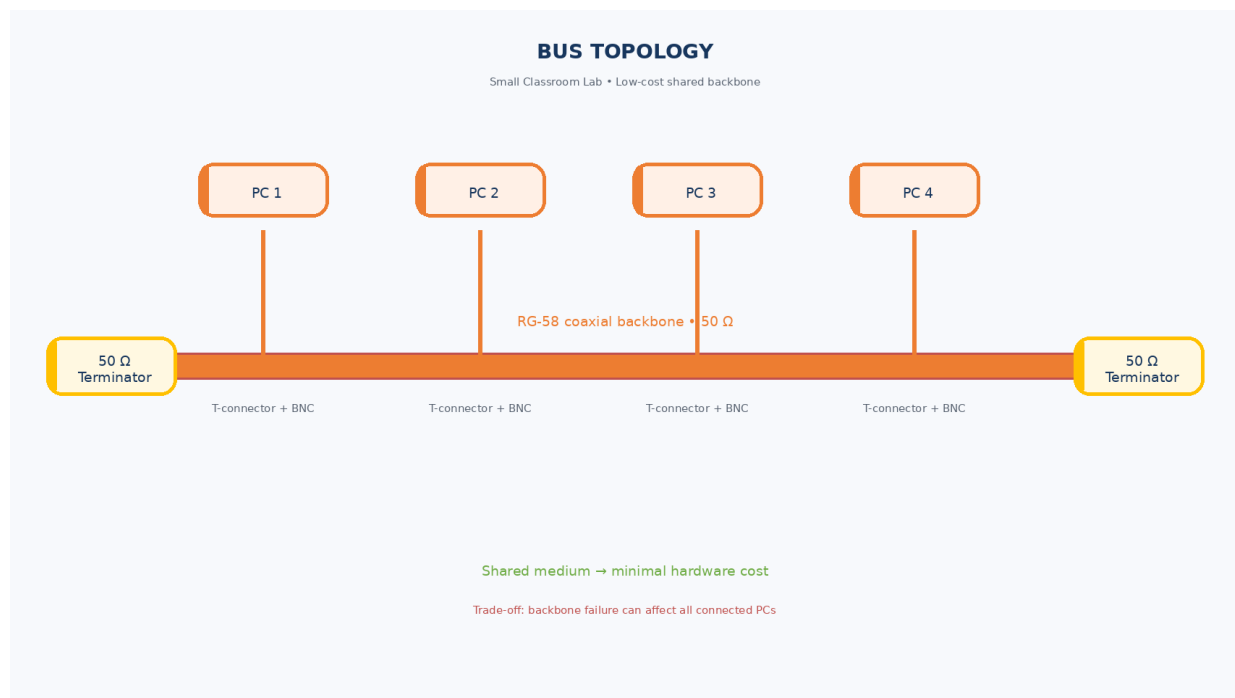


Figure 3: Bus topology for the classroom laboratory

- Classic implementation: RG-58, 50-ohm coaxial backbone.
- Each computer connects using a T-connector and BNC connector.
- 50-ohm terminators are placed at both ends of the backbone to absorb signals and prevent reflection.

- The source also describes a contemporary low-cost alternative using Cat5e/Cat6 with an inexpensive unmanaged switch for bus-like simplicity.

Technical Implementation

1. Lay the shared backbone across the classroom.
2. Attach each workstation to the backbone using the specified connector arrangement.
3. Install terminators at both ends in the classic coaxial implementation.
4. Configure basic file sharing without requiring a dedicated server.
5. Document the cable path so a physical fault can be located quickly.

Testing & Validation

- Test communication between multiple classroom PCs.
- Verify that both backbone ends are terminated correctly in the classic setup.
- Simulate a backbone break and observe the impact on connected nodes.
- Check file sharing between teacher and student PCs.

Expected result: A low-cost lab network appropriate for light usage, while clearly documenting the trade-off in fault tolerance.

Question 4 – Hybrid Topology for a University Campus

A large university campus serves thousands of users across multiple buildings — lecture halls, labs, libraries, dormitories, and administration — making a single topology inadequate. A hybrid topology combines the best elements of multiple designs: a fiber-based ring or mesh at the campus backbone level for redundancy, a star topology within each building for manageability, and bus or tree structures within individual labs for cost-efficiency. At the campus core, high-speed fiber optic cables (single-mode, 10 Gbps or 40 Gbps) interconnect core routers and multilayer switches using a ring or partial mesh arrangement, ensuring that no single link failure isolates an entire building. Each building has its own distribution switch connected to the campus core via dual fiber uplinks. Within each building, a traditional star topology branches from the distribution switch to floor-level access switches, which then connect individual devices via Cat6 cables. This hierarchical three-layer model — core, distribution, and access — is the industry standard for enterprise and campus networks because adding a new building simply means installing a new distribution switch and running fiber to the core, with no disruption to existing infrastructure. Wireless access points throughout the campus connect via PoE (Power over Ethernet) to access switches, eliminating the need for separate power cabling. The hybrid design thus delivers redundancy at the backbone, simplicity at the building level, and scalability campus-wide.

Problem context: A university serves thousands of users across lecture halls, laboratories, libraries, dormitories and administration. A single topology cannot efficiently satisfy all campus requirements.

Problem Analysis

- The campus needs high availability at the backbone and manageable networks inside buildings.
- The design must scale when new buildings are added.
- Different areas have different cost and performance requirements.
- Wireless access must be distributed across buildings without requiring separate power cabling.

Proposed Solution Design

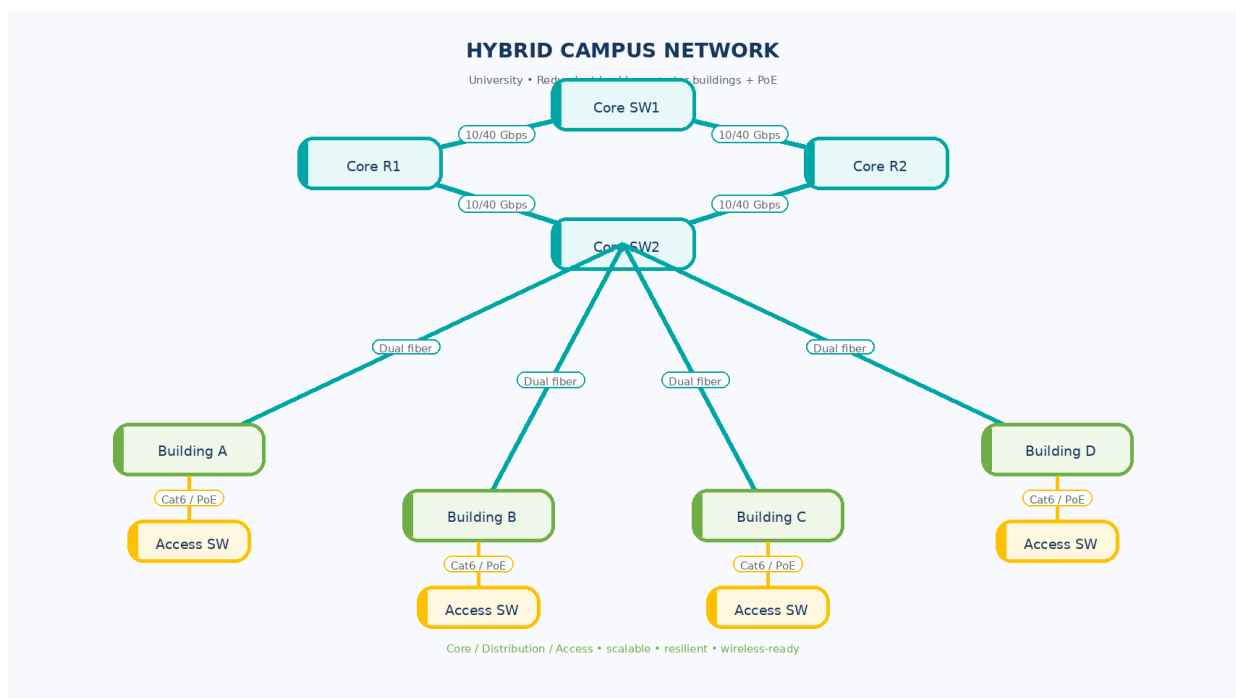


Figure 4: Hybrid topology for the university campus

- Campus backbone: fiber-based ring or partial mesh for redundancy.
- Building networks: star topology for manageability.

- Labs may use bus/tree structures where cost efficiency is required.
- Core links: high-speed fiber; the source specifies single-mode links and 10/40 Gbps examples.
- Each building uses a distribution switch with dual fiber uplinks to the campus core.
- Access switches connect endpoints using Cat6 cables.
- Wireless access points use PoE from access switches.

Technical Implementation

1. Deploy a three-layer hierarchy: core, distribution and access.
2. Connect building distribution switches to the core using dual fiber links.
3. Connect floor-level access switches to the building distribution switch.
4. Connect end devices using Cat6 cabling.
5. Deploy PoE-capable access switches for wireless access points.
6. Add a new building by extending fiber to the core and installing a new distribution switch.

Testing & Validation

- Fail one backbone link and verify the building remains reachable through the alternate path.
- Test connectivity between buildings.
- Verify wired and wireless client access.
- Add a simulated new building and verify that existing services remain operational.

Expected result: A scalable campus architecture combining backbone redundancy, building-level manageability and cost-efficient local structures.

Question 5 – Departmental Hierarchical Network with VLANs

Designing a network for a company with multiple departments involves creating a structured and scalable architecture using switches and routers to ensure efficient communication and security. Each department (such as HR, Finance, Sales, and IT) is connected through dedicated Layer 2 switches that form the access layer, allowing devices like computers and printers to communicate within the same department. These switches are then connected to a central Layer 3 switch or router, which acts as the core layer and enables communication between different departments through inter-VLAN routing. Virtual LANs (VLANs) are configured to logically separate each department's network, improving security and reducing broadcast traffic. A router is also used to connect the internal network to external networks such as the internet, often with firewall functionality for security. Redundant links and backup paths can be included to ensure reliability, while proper IP addressing and subnetting are applied for efficient network management. This hierarchical design improves performance, enhances security, and allows easy expansion as the company grows.

Problem context: A company with HR, Finance, Sales and IT departments needs efficient communication, security, controlled broadcasts and easy expansion.

Problem Analysis

- Departments need logical separation even when they share physical infrastructure.
- Inter-department communication must be controlled and routed.
- Broadcast traffic should be reduced through segmentation.
- The design must remain scalable as the company grows.

Proposed Solution Design

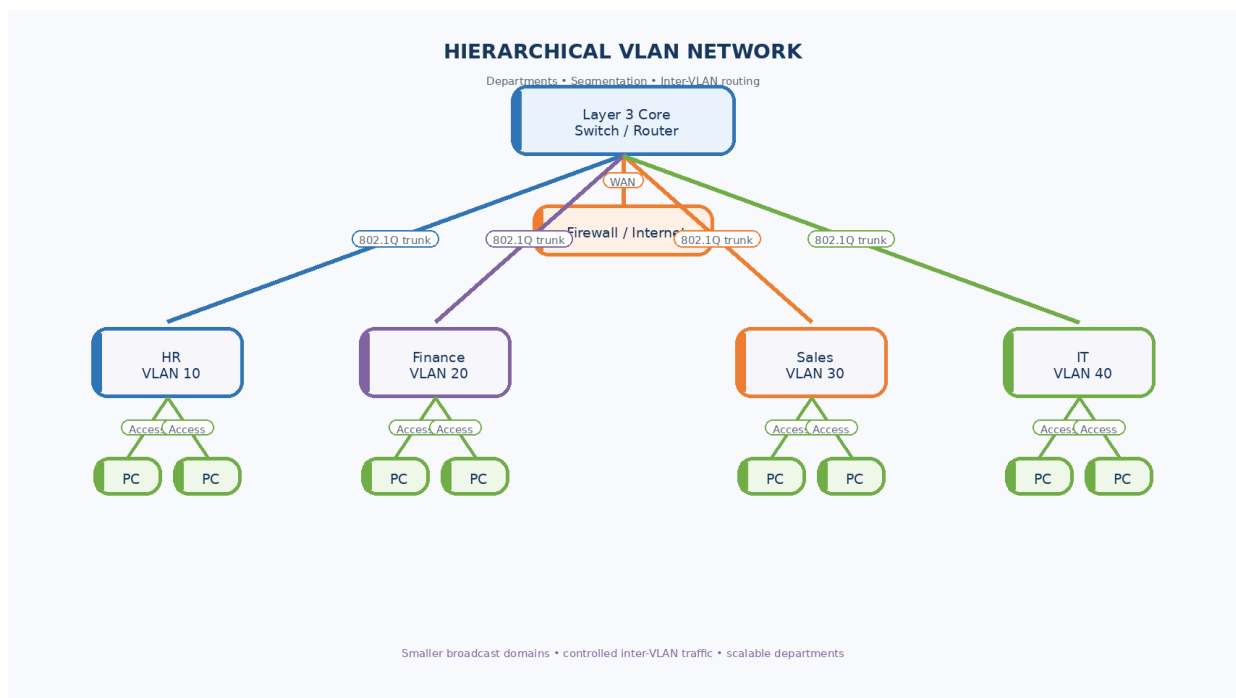


Figure 5: Hierarchical departmental network with VLAN segmentation

- Access layer: dedicated Layer 2 switches for each department.
- Core layer: central Layer 3 switch or router.
- Logical segmentation: separate VLANs for HR, Finance, Sales and IT.
- Inter-VLAN routing: performed by the Layer 3 core device.

- Internet edge: router/firewall provides external connectivity and security.
- Redundant links and backup paths can be included for reliability.
- Proper IP addressing and subnetting support efficient management.

Technical Implementation

1. Create a VLAN for each department and assign department devices to the appropriate VLAN.
2. Connect departmental Layer 2 switches to the Layer 3 core.
3. Configure inter-VLAN routing on the core device.
4. Apply appropriate IP addressing and subnetting.
5. Connect the core to a router/firewall for Internet access.
6. Use redundant links where required for availability.

Testing & Validation

- Verify devices within each department can communicate as intended.
- Verify inter-VLAN communication through the Layer 3 device.
- Check that broadcast traffic is contained within VLAN boundaries.
- Test Internet access through the router/firewall.
- Disconnect one redundant link, where implemented, and verify continued connectivity.

Expected result: A secure, segmented and scalable hierarchical network that improves performance and simplifies future expansion.